



Vaxcel®

HEPARIN SODIUM

INJECTION 5000 I.U./ml

COMPOSITION:

Each vial contains
Heparin Sodium (Bovine Mucosa)
25000 I.U./5ml

1ml of solution contains

Heparin Sodium (Bovine Mucosa) 5000 I.U.
Benzyl Alcohol 10mg as preservative.

PRESENTATION:

Sterile, clear solution.

INDICATIONS:

Prophylaxis and treatment of venous thrombosis and pulmonary embolism;
Treatment of myocardial infarction and arterial embolism;
For prevention of clotting in arterial and heart surgery and prevention of cerebral thrombosis;
As an anticoagulant in blood transfusions, extracorporeal circulation, dialysis and for lab purposes.

PHARMACODYNAMIC:

Heparin inhibits reactions that lead to the clotting of blood and the formation of fibrin clots *in vitro* and *in vivo*. Heparin acts at multiple sites in the normal coagulation system. Small amounts of heparin in combination with anti-thrombin III (heparin cofactor) can inhibit thrombosis by inactivating activated Factor X and inhibiting the conversion of prothrombin to thrombin. Clotting time is prolonged by full therapeutic doses of heparin; in most cases, it is not measurably affected by low doses of heparin. Peak plasma levels of heparin are achieved 2 to 4 hours following subcutaneous administration, although there are considerable individual variations. Heparin does not have fibrinolytic activity, therefore it will not lyse existing clots.

PHARMACOKINETICS:

Heparin is not absorbed from the gastrointestinal tract. It is extensively bound to plasma proteins following intravenous or subcutaneous injection. It does not cross the placenta and not excreted in breast milk. The half-life of heparin depends on the dose and

route of administration as well as the method of calculation and is subject to wide inter- and intra-individual variation; a range of 1 to 6 hours with an average of 1.5 hours. It may be slightly prolonged in renal impairment, decreased in patients with pulmonary embolism, and either increased or decreased in patients with liver disorder. Heparin is taken up by the reticuloendothelial system. It is excreted in the urine, mainly as metabolites, although, following administration of large doses, up to 50% may be excreted unchanged.

RECOMMENDED DOSAGE:

Intravenous administration : 5,000 - 10,000 i.u. every four hours either by bolus injection or continuous infusion in Sodium Chloride Injection or Dextrose Injection. However, the dose should be monitored with coagulation tests performed just before each administration and varied according to individual response. The clotting time should be 2-3 times the control value.

Subcutaneous administration (Therapeutic dosage) : Subcutaneous administration of 10,000 i.u. may be given every 8 hours after an initial intravenous bolus injection of 5,000 i.u.

Low-dose heparin prophylaxis : 5,000 i.u. should be given two to six hours pre-operatively and every 8 - 12 hours post-operatively for 10 - 14 days, or until the patient is mobile, whichever is the longer.

Haemodialysis : The dosage must be determined individually, depending on the patient's coagulations status and the type of apparatus used.

Myocardial infarction : 5,000 i.u. s.c. every twelve hours beginning during the twelve hours following the first sign of myocardial infarction.

Open heart surgery : Operations of less than two hours, 120 i.u./kg/hour. For operations of longer duration, one and a half times this dose should be given. For each 450ml of blood used 2,000 i.u. are needed.

Treatment periods vary from 10 - 14 days in perioperative prophylaxis to as much as six weeks in the treatment of established thrombosis. It is anticipated that heparin will have disappeared from the blood stream 4 hours after intravenous injection of 5,000 i.u. and 6 - 8 hours after 10,000 i.u. and 15,000 i.u. of i.v. heparin, respectively.

The intramuscular route cannot be recommended because of high incidence of haematoma. The increase in clotting time provided by heparin becomes apparent immediately after administration and last for 4 to 6 hours after intravenous injection and for about eight hours after subcutaneous injection. Heparin without preservatives should be used in premature infants.

Dosage in the elderly:

Elderly women have a greater tendency to bleed and it may be necessary to reduce the dose according to coagulant tests, but dosage alterations are unlikely for prophylaxis.

ROUTE OF ADMINISTRATION:

Administered by subcutaneous or intravenous injection or by intravenous infusion after dilution with a suitable vehicle solution.

CONTRAINDICATIONS:

Heparin is contraindicated in patients known to have hypersensitivity to heparin. It is also contraindicated when suitable blood coagulation tests - e.g. the whole-blood clotting time, partial thromboplastin time, - cannot be performed at the required intervals. There is usually no need to monitor the effect of low dose heparin in patients with normal coagulation parameters. The drug is contraindicated during any uncontrolled active bleeding state (see Warnings). Heparin without preservative should be used in premature infants.

SIDE EFFECTS:

Transient alopecia and diarrhoea may occur. Thrombocytopenia and osteoporosis with spontaneous fractures have been reported. Febrile or allergic reactions have occasionally been reported.

SYMPTOMS AND TREATMENTS FOR OVERDOSAGE:

Bleeding may be a complication of therapy. Slight epistaxis, occasional red cells in the urine, and bruising are signs of overdosage. Slight haemorrhage due to overdosage can usually be treated by withdrawing the drug. Severe bleeding may be reduced by the administration of protamine sulphate. The effect of heparin can be reversed immediately by intravenous administration of 1% protamine sulphate solution. The injection should be given very slowly (over one to three minutes). The quantity of protamine required for neutralization falls rapidly with the lapse of time after administration of heparin. If given within 15 minutes of the heparin injection 10mg of protamine will neutralize 1,000 i.u. of heparin, while 30 minutes after the heparin injection of 1,000 i.u., only 5mg of protamine sulphate is needed. If more time has elapsed after the administration of heparin, the dose of protamine sulphate required for neutralization should be determined accurately by titrating with the patient's plasma. It is important to avoid overdosage of protamine sulphate because protamine itself has anticoagulant properties. The dosage should not exceed the equivalent of 50mg protamine sulphate in any ten-minute period. Intravenous injection of protamine may cause a sudden fall in blood pressure, bradycardia, dyspnoea, and transitory flushing, but these may be avoided or diminished by slow administration.

WARNING:

When heparin sodium is administered in therapeutic amounts, its dosage should be regulated by frequent blood coagulation tests. If the coagulation test is unduly prolonged or if haemorrhage occurs heparin sodium should be discontinued promptly (See Overdosage). Some of the condition in which increased danger of haemorrhage exists are as follows :

Cardiovascular - Subacute bacterial endocarditis; arterial sclerosis; increased capillary permeability; during and immediately following (a) spinal tap or spinal anaesthesia or (b) major surgery, especially involving the brain, spinal cord or eye.

Haematologic - Conditions associated with increased bleeding tendencies, such as haemophilia, some purpuras, and thrombocytopenia.

Gastrointestinal inaccessible ulcerative lesions and continuous tube drainage of the stomach or small intestine. Heparin may prolong the one-stage prothrombin time. Therefore, when heparin sodium is given with dicumarol of warfarin sodium, a period of at least five hours after the last intravenous dose or 24 hours after the last subcutaneous dose should elapse before blood is drawn if a valid prothrombin time is to be obtained.

As this preparation contains benzyl alcohol, its use should be avoided in children under 2 years of age. Not to be used in neonates.

PRECAUTIONS:

Heparin therapy should be given with caution to patients about to undergo surgery and those with impaired renal or hepatic function. If oral anticoagulants are started, heparin should be continued in slightly decreasing doses for another 4 - 5 days until the oral drug has attained full prothrombin depressing activity. Heparin should be used with caution in any patient with a history of allergy. Before a therapeutic dose is given to such a patient, a trial dose of 1,000 units may be advisable.

PREGNANCY AND LACTATIONS:

The antithrombotic drug of choice during pregnancy should be heparin, even taking into consideration the fact that long-term (6 months or more) application of heparin can cause severe osteoporosis in the mother. To minimize the risk of osteoporosis heparin is given in the first trimester, following by coumarin therapy until about the 36th week, and then heparin is given for the last few weeks. Heparin therapy should be completely stopped six hours before delivery. Heparin does not cross the placental barrier and is not secreted into breast milk.

DRUG INTERACTIONS:

Drugs (such as acetylsalicylic acid, dextran, phenylbutazone, ibuprofen, indomethacin, dipyridamole, and hydroxychloroquine) that interfere with platelet-aggregation reactions may induce bleeding and should be used with caution in patients receiving heparin. It may be necessary to increase doses of heparin in the febrile state. Digitalis, tetracycline, nicotine, or antihistamines may partially counteract the anticoagulant action of heparin sodium. An increased resistance to the drug is frequently encountered in thrombosis, thrombophlebitis, infection with thrombosing tendencies, myocardial infarction, cancer, and postsurgical patients.

STORAGE:

Store below 30°C. If withdrawal of injection solution from the container is performed under aseptic conditions, a vial can be stored over up to 14 days after the first use. The date of first withdrawal must be noted on the label. Not to be used if solution is not clear.

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KEEP OUT OF REACH OF CHILDREN
JAUHI DARI KANAK-KANAK

PACKAGE QUANTITIES:

Available in vial 5ml x 1's & 5ml x 10's.
Not all pack sizes may be marketed.

Further information can be obtained from pharmacist,
physician or the manufacturer.